



Lab 05: Enhance Application Reliability with Internal Load Balancing

Build a Secure Google Cloud Network | GSP216

Overview

Set up an Internal Load Balancer using two managed instance groups in the same region but different zones. The ILB distributes TCP traffic across private backends without exposing them to the public internet. This mimics a real-world microservice or API endpoint that only needs to be accessible internally.

Internal vs External Load Balancer

Internal Load Balancer	External Load Balancer
Private IP only	Public IP accessible from internet
Regional service	Global or regional
Used for microservices and APIs	Used for public-facing web apps
Not exposed to internet	Exposed to internet

Objectives

- Configure HTTP and health check firewall rules for internal backends
- Create instance templates for consistent VM deployments
- Deploy managed instance groups in two zones for high availability
- Configure an Internal Load Balancer with a static private IP
- Test traffic distribution using curl from a utility VM

Task 1 - Configure Firewall Rules

Created two firewall rules for the my-internal-app network targeting VMs with the lb-backend tag.

- app-allow-http: TCP port 80 from 10.10.0.0/16 targeting lb-backend tag
- app-allow-health-check: TCP from 130.211.0.0/22 and 35.191.0.0/16 targeting lb-backend tag

allow-ssn					
☒	app-allow-health-check	Ingress	lb-backend	IP ranges:	tcp
☒	app-allow-http	Ingress	lb-backend	IP ranges:	tcp:8
☐	app-allow-	Ingress	Apply to all	IP ranges:	icm

Task 2 - Configure Instance Templates and Groups

Created two instance templates (one per subnet) and deployed managed instance groups in different zones for zone-level redundancy.

- instance-template-1: subnet-a, lb-backend tag, Apache PHP startup script
- instance-template-2: subnet-b, copied from template 1
- instance-group-1: Zone A, 1 instance, 80% CPU autoscale
- instance-group-2: Zone B, 1 instance, 80% CPU autoscale
- utility-vm: subnet-a, static IP 10.10.20.50, used for testing

```
curl 10.10.20.2
curl 10.10.30.2
```

VM instances

Create instance

Import VM

Refresh

Learn

Instances

Observability

Instance schedules

VM instances

Filter Enter property name or value

Status	Name ↑	Zone	Connect
☒	instance-group-1-5m33	europe-west1-d	SSH
☐	instance-group-2-285x	europe-west1-b	SSH

Related actions

Fix data protection

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ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-04-2a76ed65c66c/zones/europe-west1-d/instances/utility-vm?authuser=0&hl=...

SSH-in-browser

UPLOAD FILE

DOWNLOAD FILE

Linux utility-vm 6.1.0-49-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64

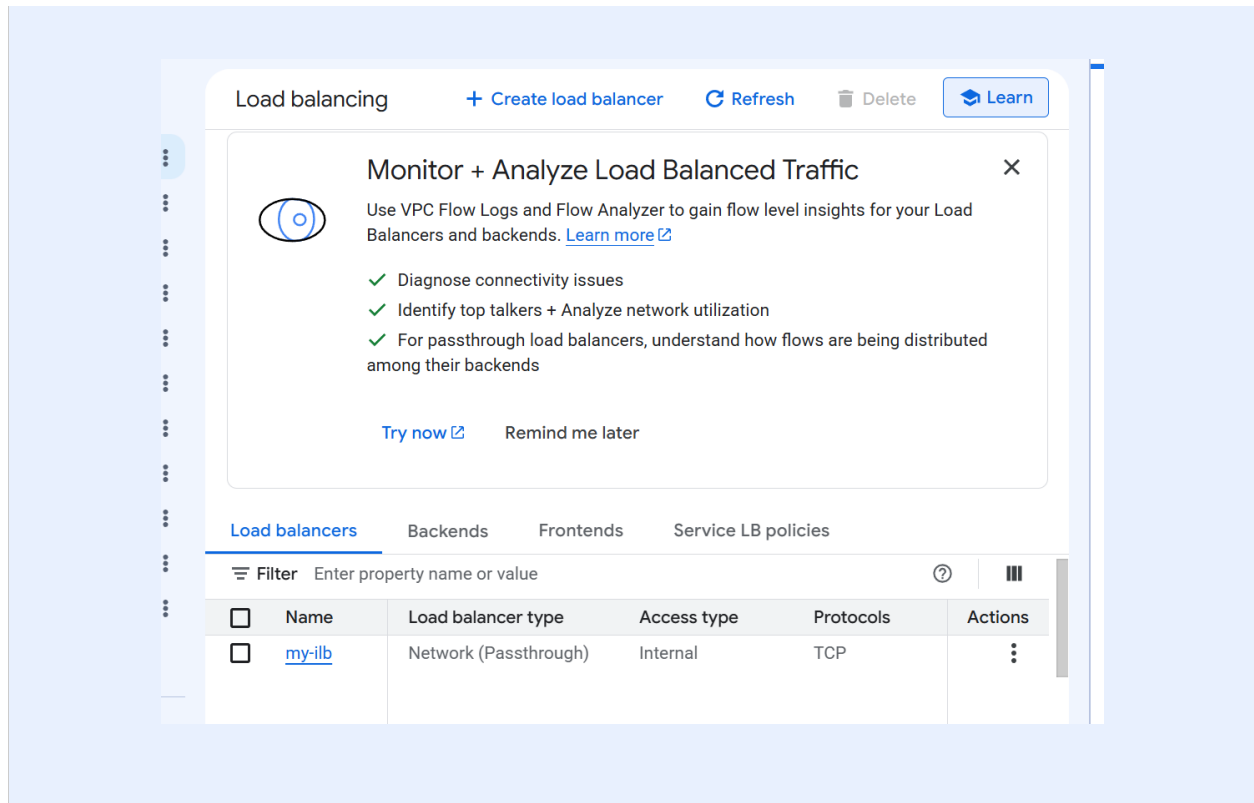
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Creating directory '/home/student-00-5c42db87d7bf'.
student-00-5c42db87d7bf@utility-vm:~\$ curl 10.10.20.2
<h1>Internal Load Balancing Lab</h1>
<h2>Client IP</h2>
Your IP address : 10.10.20.50<h2>Hostname</h2>
Server Hostname: instance-group-1-5m33<h2>Server Location</h2>
Region and Zone: student-00-5c42db87d7bf@utility-vm:~\$ curl 10.10.30.2
<h1>Internal Load Balancing Lab</h1>
<h2>Client IP</h2>
Your IP address : 10.10.20.50<h2>Hostname</h2>
Server Hostname: instance-group-2-285x<h2>Server Location</h2>
Region and Zone: student-00-5c42db87d7bf@utility-vm:~\$

Task 3 - Configure the Internal Load Balancer

Created an Internal Load Balancer named my-ilb with a static private IP of 10.10.30.5 in subnet-b.

- Type: Network Load Balancer, Passthrough, Internal
- Region: same region as both instance groups
- Network: my-internal-app
- Backends: instance-group-1 and instance-group-2
- Health check: TCP port 80, named my-ilb-health-check
- Frontend IP: 10.10.30.5 (static, reserved as my-ilb-ip)
- Port: 80

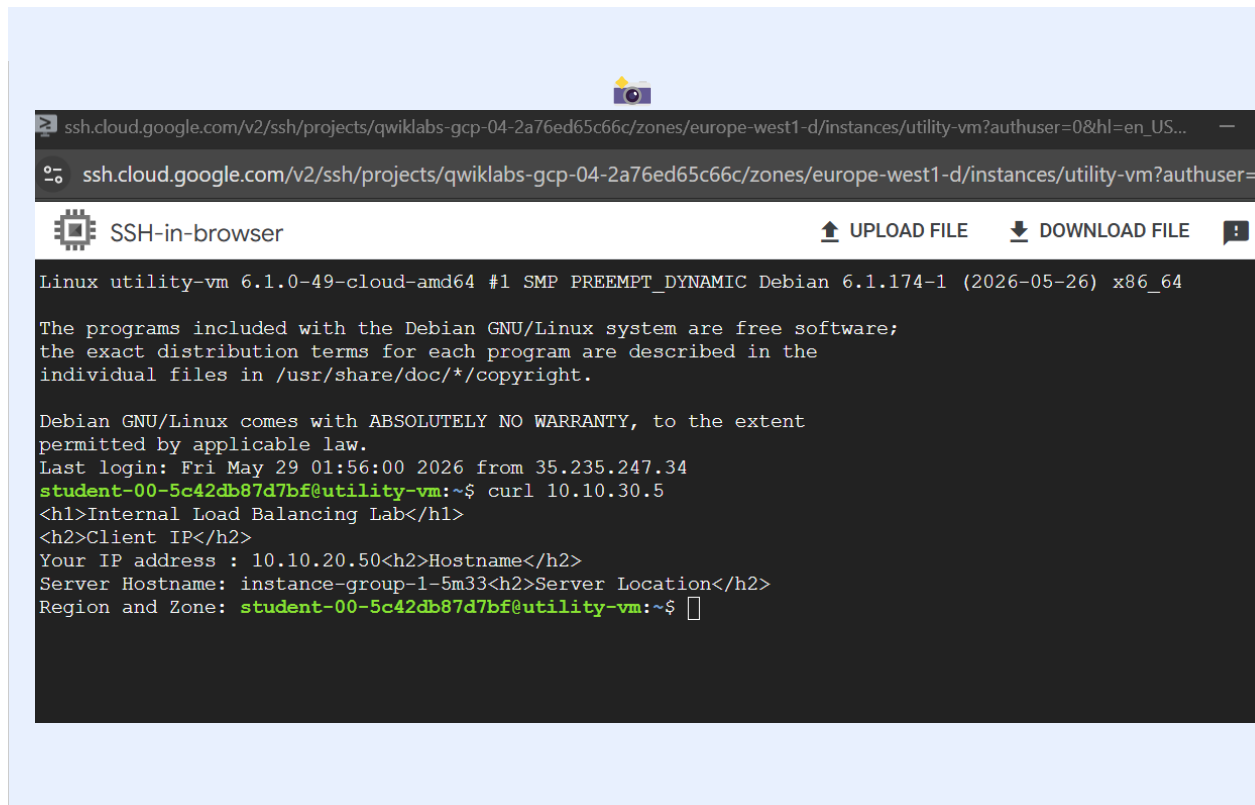


Task 4 - Test the Internal Load Balancer

SSH'd into utility-vm and ran curl against the ILB IP multiple times to verify traffic is distributed across both backends.

```
curl 10.10.30.5
```

Results: Responses alternated between instance-group-1 in Zone A and instance-group-2 in Zone B confirming traffic distribution is working.



```
ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-04-2a76ed65c66c/zones/europe-west1-d/instances/utility-vm?authuser=0&hl=en_US...
ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-04-2a76ed65c66c/zones/europe-west1-d/instances/utility-vm?authuser=
SSH-in-browser
Linux utility-vm 6.1.0-49-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Fri May 29 01:56:00 2026 from 35.235.247.34
student-00-5c42db87d7bf@utility-vm:~$ curl 10.10.30.5
<h1>Internal Load Balancing Lab</h1>
<h2>Client IP</h2>
Your IP address : 10.10.20.50<h2>Hostname</h2>
Server Hostname: instance-group-1-5m33<h2>Server Location</h2>
Region and Zone: student-00-5c42db87d7bf@utility-vm:~$
```

Real World Application

- Microservices use internal load balancers to talk to each other without public exposure
- Database clusters use ILB to distribute read queries across replicas
- Multi-zone deployment with ILB ensures the app stays up even if one zone fails
- Private APIs are accessible internally without being exposed to the internet

What I Learned

- Internal Load Balancers use private IPs and are only accessible within the VPC
- Deploying instance groups in different zones provides zone-level redundancy
- Health checks ensure traffic only goes to healthy backend instances
- Instance templates make it easy to replicate consistent VM configurations
- The lb-backend network tag connects instances to the correct firewall rules
- A utility VM inside the same network is needed to test internal resources

Resources

GCP Internal Load Balancing Docs: <https://cloud.google.com/load-balancing/docs/internal>
Lab ID: GSP216